

SYED MAOZ ARIF

Hardware Developer & Electronics Engineer

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Abbottabad, Pakistan

PROFESSIONAL SUMMARY

Hardware developer and electronics engineering student (5th semester) at COMSATS University with hands-on experience in multi-layer PCB design, embedded systems, firmware development, and IoT. Builder of 12 real-world projects, holder of 17 certifications, and winner of 5 awards including a global Instructables contest. Active freelancer serving international clients. Proven leadership abilities as President of IET On Campus.

EDUCATION

B.S. Electrical Engineering (Focus: Electronics)

COMSATS University Islamabad, Abbottabad Campus

2024 – 2028 (Expected)

5th Semester / GPA: 3.61/4.0

- **Focus Areas:** Embedded Systems, PCB Design, Signal Integrity, IoT, Firmware Engineering, Digital Logic
- **Honors:** Altium Global Scholarship Awardee (March 2026)

TECHNICAL SKILLS

- **PCB & Hardware Design:** Altium Designer, Altium 365, CircuitStudio, KiCad, EasyEDA, Proteus, LTspice, Multisim, Schematic Capture, Impedance Matching, Signal Integrity
- **Embedded Systems & IoT:** ESP32, STM32, Arduino, ATMEGA, Lattice FPGA, I2C, SPI, UART, RF, MQTT, Firebase, Digital Logic
- **Programming & Firmware:** C/C++ (Embedded C), Python, Verilog (HDL), MATLAB, JavaScript, HTML5, CSS3
- **3D CAD & Modeling:** Autodesk Inventor, SolidWorks, Fusion 360, AutoCAD, 3D Printing, Enclosure Design

PROFESSIONAL EXPERIENCE & LEADERSHIP

Freelance Hardware & Mechanical Designer

Fiverr & Upwork

January 2024 – Present

Remote

- Deliver professional multi-layer PCB designs, schematic capture, and 3D CAD modeling for international clients.
- Manage project scoping, component selection (BOM), deadline management, and technical documentation.

President (Previously VP & Membership Head)

IET On Campus, CUI Abbottabad

May 2025 – Present

Abbottabad, PK

- Guide chapter operations and represent the student branch to foster engineering excellence for 50+ members.
- Organize technical events, professional development initiatives, and specialized workshops (e.g., AI in UAVs).

Cabinet Member

Youth Parliament Pakistan

2025 – 2027

National

- Serve in a leadership capacity for youth advocacy, engineering sector representation, and national policy discussions.

KEY PROJECTS

Industrial Edge AI Condition Monitoring & 3D Digital Twin

STM32H7, ESP32 WROOM, Ethernet, Altium Designer, C/C++

July 2026

- Designed an industrial-grade development board with dual-MCU architecture, hardware calibration, and 70 GPIOs.
- Secured 1st Position (Software Category) at CEP Exhibition SP26 and 2nd Position at EMCOT 2026.

Stratum-1 Precision Hardware Clock & GPS Bridge

Verilog, Lattice iCE40UP5K FPGA, NEO-6M GPS, Digital Logic

August 2026

- Engineered a Stratum-1 time synchronizer using pure digital logic to bypass microprocessor latency entirely.
- Developed a custom hardware UART receiver and NMEA state machine to parse GPS signals directly in silicon.

ElectroParts IMS: Inventory Architecture

JavaScript, Firebase, CapacitorJS, HTML5/CSS3

August 2026

- Architected a cross-platform (Web/Native Android) inventory engine supporting 32 electronics categories.
- Implemented dual-storage local/cloud sync and linear barcode generation algorithms for zero-latency retrieval.

IncliSense Precision Digital Level

Arduino Nano, MPU6050, IoT, C++, 3D Printing

April 2026

- Built a precision leveling device analyzing dual-axis spatial data via I2C, yielding high-accuracy PC telemetry.
- Awarded Global Runner-Up in the Instructables Sensors Contest.

Mixed-Signal RF Transceiver & 6S UAV BMS

2026

ESP32, RF Routing, BQ76930, Altium Designer

- Designed an ESP32 PCB with 50-ohm impedance-matched RF traces and 312 stitching vias for a Faraday cage.
- Developed a 6-cell battery management system schematic ensuring high-current safety via MOSFET protections.

AWARDS & HONORS

- **1st Position, Best CEP Project Award (Software Category)** - COMSATS University & District Youth (July 2026)
- **2nd Position, EMCOT 2026 Best Complex Engineering Project** - COMSATS & IET (August 2026)
- **Runner-Up, Global Sensors Contest** - Instructables (April 2026)
- **Altium Global Scholarship Awardee** - Altium Education (March 2026)
- **1st Position, Best Circuit Design** - COMSATS University & IET Hardware Workshop (Feb 2026)

CERTIFICATIONS

- Altium Certified: Designer Essentials | Altium Education PCB Basic Design | Generative AI Mastermind
- Arduino & IoT (Cardiff Institute) | 3D Printing Certificate | Solar Solution for Pakistan
- Additional 11 certifications in Leadership, AI, and Engineering Project Management.